

- 5 (a) $\frac{1}{2}mv^2 = qV$ (or some verbal explanation) B1
- $\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = 1.6 \times 10^{-19} \times 1.2 \times 10^4$ B1
- $v = 6.49 \times 10^7 \text{ m s}^{-1}$ A0 [2]
- (b)(i) *within field:* circular arc B1
- in 'downward' direction B1
- beyond field:* straight, with no 'kink' on leaving field B1 [3]
- (ii) 1. v is smaller M1
- deflection is larger A1 [2]
2. (magnetic) force is larger M1
- deflection is larger A1 [2]